Lake Ontario and Yellow Creek Ravine: finite mathematics for qubit-assisted robotic monitoring and ten-year remediation scenarios

Independent research preprint | Nature-style article structure

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Abstract

We formulate a finite-horizon framework connecting hypothetical superconducting qubit-assisted spectral analysis, autonomous water sampling, and watershed remediation scenarios. Transmon, coplanar-waveguide, fluid-flow, natural-gradient and discrete stability equations describe the proposed architecture. Numerical tables and plots are generated directly from the application's existing recovery model. At elapsed year 10, modeled phosphorus loading is 6,540.5 tonnes/year and Yellow Creek E. coli is 148.1 CFU/100 mL, not the asymptotic targets of 6,200 and 78. Hardware performance, quantum advantage, hydraulic feasibility and regulatory compliance remain unvalidated.

1. Model provenance and finite horizon

The data source is OntarioQubitSpectroscopyEngine.compute_lake_ontario_10yr_recovery. The model uses ten annual samples n = 1,...,10. Its labels span 2026-2035; a baseline at n = 0 is mathematically separate. A literal 2026-2036 interval would need an explicitly defined baseline and calendar convention. No measured time series, fitted parameters, confidence intervals or intervention trials are supplied by this model.

$$x_n = x^* + (x_0 - x^*) \exp(-k\, n),\ n \in \{0,...,10\}. \quad (1)$$

Equation (1) implies x(n) approaches x* asymptotically; the target is not attained at a finite year merely because it appears in the formula.

2. Qubit circuitry and finite-dimensional readout

$$H = 4E_C(N - n_g I)^2 - E_J \cos(\Phi) + \hbar \omega_r a^\dagger a + \hbar \chi\, a^\dagger a\, \sigma_z. \quad (2)$$

N and Phi denote charge and phase operators. A numerical implementation must specify a finite charge basis and a resonator photon cutoff. For six two-level qubits the register dimension is $2^6 = 64$; this dimension alone does not imply a 64-fold computational advantage.

$$\varepsilon_m \approx (-1)^m E_C \frac{2^{4m+5}}{m!} \sqrt{\frac{2}{\pi}} \left(\frac{E_J}{2E_C}\right)^{m/2+3/4} \exp\left[-\sqrt{\frac{8E_J}{E_C}}\right]. \quad (3)$$

This asymptotic transmon charge-dispersion expression requires a large E_J/E_C ratio. It does not prove immunity to noise or a particular coherence time. The earlier listed energies are inconsistent: 248.5 GHz / 0.280 GHz = 887.5, not 88.75. Energies quoted in GHz must mean E/h; angular frequencies require omega = 2 pi f.

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 t)^{-1/2}\, dt;\ K'(k) = K(\sqrt{1-k^2}). \quad (4)$$

$$Z_0 \approx \frac{30\pi}{\sqrt{\varepsilon_{eff}}} \frac{K'(k)}{K(k)},\ k = w/(w+2s). \quad (5)$$

Equation (5) is the ideal quasi-static coplanar-waveguide convention for center width w and gap s, not a universal transmission-line formula. Substrate geometry and electromagnetic validation are required to establish 50-ohm matching. Keeping an optical water sample separate from a millikelvin processor is essential; a practical cryogenic AUV has not been demonstrated here.

3. Hydraulic formulation and robotic collection

$$\rho(\partial u/\partial t + u \cdot \nabla u) = -\nabla p + \mu \nabla^2 u + f; \quad \nabla \cdot u = 0. \quad (6)$$

$$\rho U_j \partial U_i/\partial x_j = -\partial P/\partial x_i + \partial/\partial x_j[\mu(\partial U_i/\partial x_j + \partial U_j/\partial x_i) - \rho \langle u'_i u'_j \rangle]. \quad (7)$$

$$Re = \rho U D_h/\mu; \quad \mu_t = \rho C_\mu k^2/\varepsilon. \quad (8)$$

The app assigns U = 1.45 m/s, Re = 48,500, turbulent kinetic energy k = 0.185 m^2/s^2, and bed shear 8.42 Pa. These are scenario inputs, not outputs of a mesh-based RANS solver. Channel geometry, boundary conditions, dissipation and mesh-convergence studies are missing.

$$Q_{SI} = Q_{mL/min}\, 10^{-6}/60; \quad v_{in} = Q_{SI}/A; \quad \Delta P = \rho v_{in}^2/2 + R_h Q_{SI}. \quad (9)$$

For Q = 150 mL/min and A = 10^{-4} m^2, v_{in} = 0.025 m/s and the dynamic pressure is 0.3125 Pa at ρ = 1,000 kg/m^3. The earlier 2,120 Pa claim is not supported by these SI conversions. Filter resistance must be measured.

$$Stk = \rho_p d_p^2 U/(18 \mu D); \quad \eta_{asp} = C_{in}/C_{ambient}. \quad (10)$$

Matched stream and intake velocities define ideal isokinetic sampling. A low Stokes number does not establish a universal 1.00 +/- 0.02 capture efficiency or prove absence of cell damage.

$$r_{n+1} = r_n + \Delta t\, v_n; \quad M(v_{n+1}-v_n)/\Delta t = F_{thruster} - F_{drag} + F_{buoyancy}. \quad (11)$$

Equation (11) supplies a finite-step robot model to be calibrated. The existing intake animation is illustrative, not a validated navigation solver.

4. Natural-gradient spectral inversion

$$g_{ij} = \mathrm{Re}[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle]; \quad \theta_{n+1} = \theta_n - \eta (g + \lambda I)^{-1} \nabla L. \quad (12)$$

Here lambda > 0 regularizes the finite metric matrix. A positive metric eigenvalue alone does not prevent barren plateaus or establish a quantum speedup. The app's fidelity and speedup readouts are synthetic, without a trained circuit, held-out samples or a matched classical benchmark. The 345, 420, 582 and 680 nm peaks are synthetic optical features, not specific proof of pathogens, polymers or measured concentrations.

5. Ten-year results and conditional stability

$$P_n = 6200 + 5600 \exp(-0.28n); \quad E_n = 78 + 2322 \exp(-0.35n). \quad (13)$$

$$S_n = 12.4 - 8.6 \exp(-0.24n); \quad T_n = 2.4 + 46.1 \exp(-0.38n). \quad (14)$$

$D_n = \text{floor}[120 + 4730(1-\exp(-0.22n))]$; $I_n = 44 - 27.5\ \exp(-0.26n)$. (15)

P: phosphorus tonnes/year; E: E. coli CFU/100 mL; S: Secchi depth m; T: turbidity NTU; D: Diporeia individuals/m^2; I: illustrative benthic index. Baselines and rates are assumed, not site-calibrated.

Model metric	Year 3	Year 6	Year 10
Phosphorus (tonnes/yr)	8,617.60	7,243.70	6,540.50
Secchi depth (m)	8.21	10.36	11.62
Diporeia (individuals/m2)	2,405.00	3,586.00	4,325.00
Cladophora (g/m2)	109.20	59.10	36.60
E. coli (CFU/100 mL)	890.60	362.30	148.10
Turbidity (NTU)	17.14	7.12	3.43
Benthic index	31.40	38.20	42.00
Illustrative value (CAD M)	63.90	162.90	324.60

Table 1. Values reproduced from the current application, with its rounding and integer truncation. Asymptotic targets are not substituted for year-10 values.

Figure 1. Modeled lake-wide phosphorus trajectory

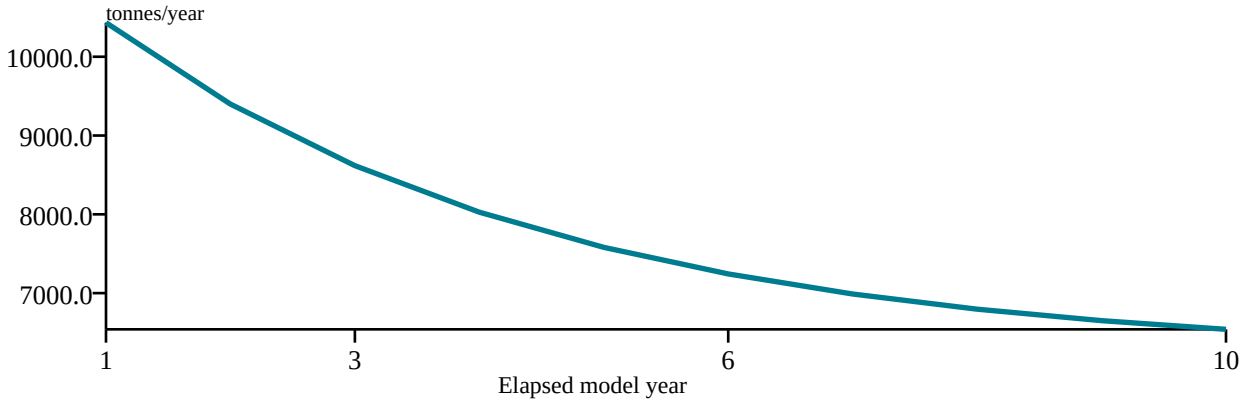


Figure 2. Modeled Yellow Creek microbial trajectory

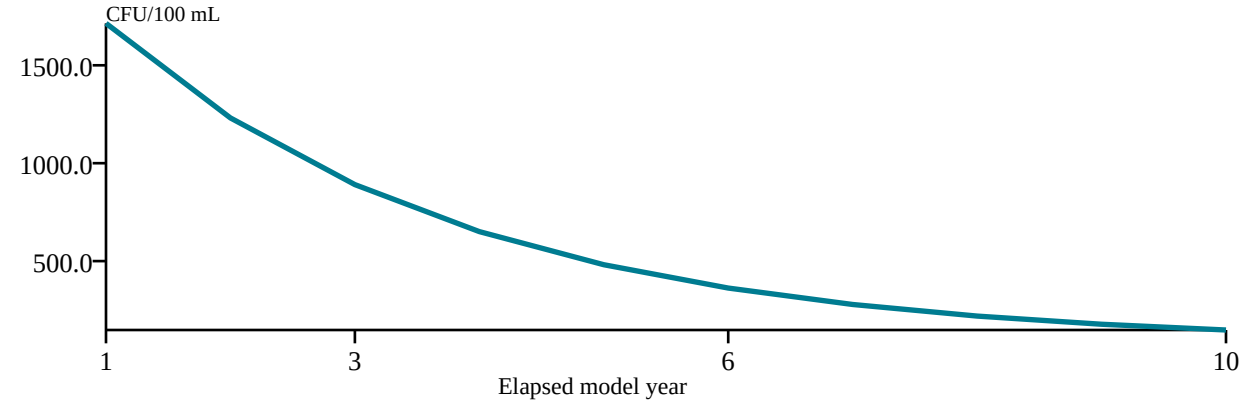
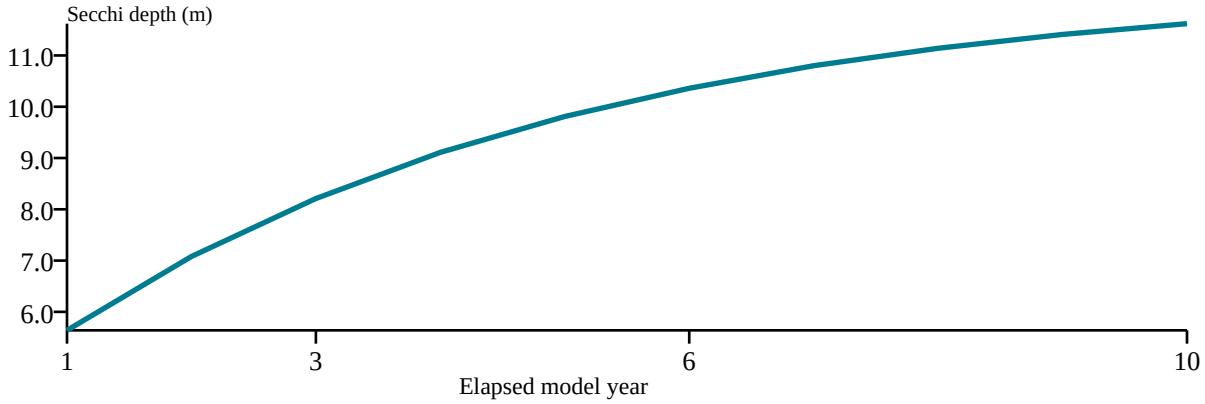


Figure 3. Modeled water transparency



$$z_{n+1} = A\, z_n,\ A = \mathrm{diag}(\exp(-0.28), \exp(-0.35));\ V_n = z_n^{T} z_n/2. \quad (16)$$

$$V_{n+1} - V_n = z_n^{T}(A^{T}A - I)z_n/2 < 0 \text{ for } z_n \neq 0. \quad (17)$$

For normalized deviations z = ((P-P*)/P*, (E-E*)/E*), Equation (17) proves convergence only for the assumed autonomous exponential model. Storm forcing, delays, saturation and ecological feedback invalidate any unconditional real-world guarantee. Year-10 outputs exceed the earlier illustrative 6,500 tonnes/year and 100 CFU/100 mL thresholds; neither these thresholds nor compliance is verified here.

6. Remediation staging and economic evaluation

Years 1-2: establish measured baselines, calibrate sampling and survey runoff hotspots. Years 3-5: evaluate source-control and riparian retention pilots under permits. Years 6-8: assess benthic response and adapt monitoring. Years 9-10: audit outcomes and sustain surveillance. Biological releases, species reintroductions and sediment interventions require independent ecological review; they are not validated deployment recommendations.

$$J = \mathrm{sum}_{n=0}^{9}[z_n^{T} W z_n + u_n^{T} R u_n], \text{ subject to state, budget and environmental constraints.} \quad (18)$$

Equation (18) defines a possible finite-horizon optimization objective; the app does not solve it, so its staging is not demonstrated optimal.

$$B(n) = 14.5\, n^{1.35} \text{ million CAD; } NPV = \mathrm{sum}_{n=1}^{10}(\mathrm{benefit}_n - \mathrm{cost}_n)/(1+r)^n. \quad (19)$$

The application produces B(10) = 324.6 million CAD. B(n) is an assumed cumulative value curve, not discounted net present value or revenue. True NPV requires incremental benefits, capital and operating costs, a discount rate, counterfactuals and uncertainty analysis.

7. Discussion and validation requirements

This manuscript specifies mathematical hypotheses and reproduces plots, rather than reporting field discoveries. Next steps include measured bathymetry and hydraulics, laboratory spectroscopy calibration, classical versus quantum benchmarking, cryogenic power and thermal feasibility, robot navigation trials, independently sourced water-quality criteria, and sensitivity and uncertainty analysis of remediation and economics. The earlier claims of established quantum advantage, noise immunity, isokinetic capture and guaranteed ecological compliance are unsupported.

Reproducibility and methodological references

Local source: backend/ontario_qubit_spectroscopy_engine.py. This PDF's generator reads its recovery arrays without modifying them. Equations (2)-(5) are idealized circuit descriptions, (6)-(11) fluid/robot formulations, (12) a proposed learning update, and (13)-(19) finite-horizon model and economic definitions.

Koch et al., Charge-insensitive qubit design derived from the Cooper pair box, Physical Review A 76, 042319 (2007), doi:10.1103/PhysRevA.76.042319. Stokes et al., Quantum Natural Gradient, Quantum 4, 269 (2020), doi:10.22331/q-2020-05-25-269. These methodological references do not validate the Lake Ontario application or its numerical assumptions.